SMART INDIA HACKATHON 2026
TITLE PAGE

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Problem Statement ID – 26047

Problem Statement Title – Patient Case-Taking Software

Theme – Smart Automation

PS Category – Software

Team ID – [Insert Team ID]

Team Name – [Insert Team Name]

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IDEA TITLE: MediKiosk

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MediKiosk — Sovereign Multimodal AI Clinical Intake & ABDM Digitization Platform

- **Autonomous Point-of-Entry Kiosk:** Deployed at OPD entry points to capture a patient's comprehensive medical history via voice + touch and digitize physical medical documents before seeing the doctor.
- **Solves 2.1-Min Consultation Crisis:** Overcomes India's severe OPD time collapse (BMJ Open, 2017) and the AYUSH Dashavidha Pariksha assessment gap by offloading structured history-taking to pre-consultation.
- **Dual Clinical Ontologies:** Integrates Western Allopathic SOCRATES framework (Site, Onset, Character, Radiation, Associations, Time, Exacerbating, Severity) and Ayurvedic Dashavidha Pariksha in a single engine.
- **Sub-150ms Emergency Red-Flag Interceptor:** Immediately flags acute cardiac, stroke, and respiratory distress symptoms for priority emergency triage rather than routine queueing.
- **100% Sovereign AI Foundation:** Powered by Sarvam AI (22 Indian languages STT, 11 languages TTS, 23 languages Document VLM)—all voice and document data processed sovereignly within India.
- **Doctor-in-the-Loop Safeguard:** Zero autonomous diagnosis—generates an editable, structured SOAP draft note allowing the physician to review complete history in 15 seconds and devote full consultation to clinical care.

1. IDENTIFY
ABHA & NHCX

→ **2. CONVERSE**
Voice Intake

→ **3. SCAN**
OCR & ABDM

→ **4. SUMMARIZE**
FHIR Bundle

→ **5. CONSULT**
15s Review

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TECHNICAL APPROACH

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1. Core Technology Stack Architecture:

- **Frontend & Client:** Flutter 3.24 (patient kiosk UI with audio streaming & touch), Next.js 15 (physician consultation dashboard).
- **Backend & State:** FastAPI (Python 3.12 async gateway), LangGraph clinical state machine, Supabase/PostgreSQL 16 with Row-Level Security (RLS).
- **Sovereign AI Foundation:** Sarvam Saaras v3 (STT, 22 Indian languages), Sarvam Bulbul v3 (TTS, 11 languages), Sarvam Vision 3B VLM (Prescription/Lab OCR, 23 languages).
- **National Health Integration:** ABDM integration via Eka Care (NHA-certified HIP/HIU gateway for M1 ABHA, M2 HIP, M3 HIU), NHCX FHIR APIs for PM-JAY eligibility, and NRCeS-compliant HL7 FHIR R4 DocumentBundle output (bdl-11 invariant).

2. Methodology & 5-Step Clinical Implementation Pipeline:

➔ **Step 1 (Identify & Consent):** ABHA QR Scan & Share + DPDPA 2023 audio consent + NHCX PM-JAY benefit check.

➔ **Step 2 (Multimodal Converse):** Adaptive voice/touch interview via Sarvam Saaras + SOCRATES/Dashavidha tree + sub-150ms red-flag triage.

➔ **Step 3 (Scan & Fetch):** Prescription OCR via Sarvam 3B VLM + Eka Care M3 HIU longitudinal health record retrieval.

➔ **Step 4 (Summarize & Route):** Structured SOAP note generation + NRCeS FHIR R4 DocumentBundle linked to hospital HIS queue.

➔ **Step 5 (15-Sec Consult):** Physician reviews pre-compiled summary, edits/confirms in 15 seconds, and devotes consultation to care.

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FEASIBILITY AND VIABILITY

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1. Feasibility Analysis & Production Viability:

- **Production-Grade Stack:** Built on existing, commercially deployed sovereign APIs (Sarvam AI, Eka Care ABDM, NHCX)—requires systems integration, not unproven fundamental research.
- **Established ABDM Pathway:** Follows standardized NHA milestone pathways (M1 → M2 → M3) supported by official reference implementations (NHA ABDM-wrapper).
- **Modest Hardware Requirements:** Operates on standard off-the-shelf kiosk hardware (touchscreen, directional noise-cancelling mic, document camera), identical to airport/ hospital check-in kiosks.

2. Potential Challenges & Risk Mitigation Strategies:

✔ **OPD Acoustic Noise (≥70dB):** Mitigation: Dual-mode voice + high-contrast icon touch interface as seamless fallback when voice input is disrupted.

✔ **Low Digital Literacy:** Mitigation: Icon-driven, audio-guided conversational UI (Sarvam Bulbul) requiring zero prior training (validated for rural/elderly).

✔ **Handwriting OCR Variations:** Mitigation: Sarvam 3B VLM trained on Indian prescriptions + mandatory doctor-in-the-loop verification before saving.

✔ **Patient Trust & Consent:** Mitigation: Audio-guided regional explanation of DPDPA consent with cryptographic logging and ephemeral zero-trace memory.

✔ **NHA Gateway Latency:** Mitigation: Graceful offline/asynchronous fallback—kiosk completes intake locally even if ABDM/NHCX times out.

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IMPACT AND BENEFITS

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1. Target Audience Impact:

- **Patients:** Elderly, rural, low-literacy, and first-visit patients currently excluded by smartphone-dependent tele-triage apps gain effortless, audio-guided OPD access.
- **Physicians:** Doctor documentation time cut from ~3 minutes to a 15-second review, redirecting saved clinical time to physical examination and counselling.
- **AYUSH Institutions:** Enables full 10-parameter Dashavidha Pariksha intake that is currently impossible to conduct manually within high-volume OPD time constraints.

2. Multi-Dimensional Benefits:

★ **Social & Clinical Quality:** Eliminates diagnostic error from rushed history-taking; ensures sub-150ms emergency triage for critical conditions.

★ **Health-System Interoperability:** Generates standardized ABDM-linked FHIR R4 EHRs, permanently solving paper record fragmentation across multiple providers.

★ **Economic Scale:** Empowers high-throughput government OPDs (4,000–10,000 patients/day) to scale intake without hiring additional triage staff.

★ **Governance & Privacy:** 100% compliant with DPDPA 2023 consent protocols, NRCeS FHIR standards, and sovereign in-country data residency.

★ **Core Transformation:** Pre-consultation history time reduced by >90% (from 3 minutes to 15 seconds).

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RESEARCH AND REFERENCES

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2. **Diagnostic Value of History-Taking:** Hampton JR, Harrison MJ, Mitchell JR, Prichard JS, Seymour C. "Relative contributions of history-taking, physical examination, and laboratory investigation to diagnosis and management of medical outpatients." BMJ. 1975;2(5969):486-489. DOI: 10.1136/bmj.2.5969.486

3. **Classical Ayurvedic Diagnostic Framework:** Charaka Samhita, Vimana Sthana, Chapter 8 (Rogabhishagjitiya Vimanam, Verses 94–130) — Standard Dashavidha Pariksha clinical ontology.

4. **National Resource Centre for EHR Standards:** NRCeS India FHIR R4 Implementation Guide for Ayushman Bharat Digital Mission (ABDM). URL: nrces.in/ndhm/fhir/r4/

5. **National Health Authority (NHA) Specifications:** ABDM Milestone 3 (M3_V1) & NHCX FHIR Coverage Eligibility Guidelines. URL: sandbox.abdm.gov.in

6. **Data Privacy & Sovereign Regulation:** The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023), Ministry of Law and Justice, Government of India. Gazette Notification Aug 11, 2023.

7. **Sovereign Indic AI Platform:** Sarvam AI Developer Documentation: Saaras v3 (STT), Bulbul v3 (TTS), Sarvam Vision 3B VLM. URL: sarvam.ai

8. **Certified National ABDM Gateway:** Eka Care ABDM Connect Developer Suite: NHA-Certified M1, M2, and M3 APIs. URL: developer.eka.care